Shared Action Errors and Rank-One Corrections for Coercive Inverse Responses

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Abstract

We study upper certificates for $\langle f, H^{-1}f\rangle$ when H is a self-adjoint operator bounded below by a positive constant and the action on a trial vector is known with error. Retaining the same error vector in the energy and residual terms gives a joint norm-ball bound, which admits a padded Gram implementation and exact strict comparisons without numerical square roots. A rank-one lower bound for the inverse slack gives two equivalent exact-action corrections and an exact complex rescaling formula; uniform interval relaxations of those corrections need not coincide. We prove a precision-only capacity bound and finite-rank Schur-complement and single-target reductions, without evaluating them. Two previously completed tests at shift $1/128$ are documented. The first excludes its fixed shared-error majorant, and its stronger common-base inequality also excludes the specified fixed-coefficient Young families. The second certifies positive rank-one corrections but remains inconclusive for the target inequality. All statements about the theta application retain the physical operator, domain, Gram, and action inclusions as conditional premises. No new scientific computation or conclusion about the Riemann hypothesis is supplied.

1 Setting and scope

Let $\mathcal{H}$ be a nonzero complex Hilbert space, with inner product conjugate-linear in the first argument. Throughout,

$$H = H^*, \qquad \mathrm{D}(H) \text{ dense in } \mathcal{H}, \qquad H \succeq aI, \quad a > 0. \tag{1.1}$$

Thus H^{-1} is a bounded positive operator on $\mathcal{H}$. For $f \in \mathcal{H}$ the quantity to be bounded is

$$\mathcal{I} = \langle f, H^{-1}f\rangle.$$

The general arguments require only (1.1) and the stated finite action information. They do not require a trial vector in $\mathrm{D}(H^2)$.

For the saved finite model, let

$$V : \mathbb{C}^{68} \longrightarrow \mathrm{D}(H), \qquad O : \mathbb{C}^{372} \longrightarrow \mathcal{H}, \qquad G, E : \mathbb{C}^{68} \longrightarrow \mathcal{H}, \qquad HV = G - hV + E, \tag{1.2}$$

where $h = 1/128$. The positive coercivity constant a is distinct from h. Fix $C \in \mathbb{C}^{68\times 372}$ and a nonzero $v \in \mathbb{C}^{372}$, and put

$$\begin{gathered} c = Cv, \quad n = v^*v, \quad t_c = c^*c, \quad f = Ov, \quad w = Vc, \\ G_h = G - hV, \quad g = G_h c, \quad r = f - g, \quad z = Ec. \end{gathered} \tag{1.3}$$

In particular $Hw = g + z$ and the true residual is $r - z$. The saved direction has 211 stored coordinates followed by 161 exact zeros. The coefficient matrix and that direction remain fixed in both reported tests. The target is exactly $\mathcal{I} < n$; there is no additional $1 - \delta$ factor.

The admitted norm and error budgets are

$$\|V\| \le \nu, \qquad E^*E \preceq \sigma^2 I, \qquad \|E\| \le \overline{\sigma}, \qquad B_i \ge \|E_i\|, \quad \sum_{i=1}^{68} B_i^2 = \sigma^2 \le \overline{\sigma}^2. \tag{1.4}$$

Here E_i is the ith error column and $\sigma \ge 0$ denotes the square root of the stored nonnegative number σ^2. The order is the original 64 columns followed by four appended columns. The triangle and Cauchy–Schwarz inequalities give

$$\|z\| \le e := \sum_{i=1}^{68} B_i |c_i| \le \sigma\sqrt{t_c} \le \overline{\sigma}\sqrt{t_c}. \tag{1.5}$$

Thus e is the smallest of these three admitted bounds. This conclusion uses the ordered sum-of-squares identity in (1.4); it is not a general comparison between unrelated error estimates. The saved real rational model makes e exact rational. This paper reuses its stored value and does not repeat the sum.

1.1 Conditional theta interpretation

To identify the inherited realization, define for $x \ge 0$

$$\kappa_\theta(x) = e^{x/2} \sum_{j \ge 1} (4z_j(x)^2 - 6z_j(x)) e^{-z_j(x)}, \qquad z_j(x) = \pi j^2 e^{2x},$$

and extend evenly. With $c_\theta(x) = \cosh(x/2)$, the inherited Hilbert space is $L^2_{\mathrm{even}}(\mathbb{R}, \nu_\theta; \mathbb{C})$, where $\mathrm{d}\nu_\theta = 2\kappa_\theta c_\theta \, \mathrm{d}x$ and $\nu_\theta(\mathbb{R}) = 1$. The symmetric half measure is

$$\mathrm{d}\mu(s) = \frac{1}{2} \frac{e^{-|s|/2}}{1 - e^{-2|s|}} \, \mathrm{d}s + \frac{1}{2} \sum_{j \ge 2} \frac{\Lambda(j)}{\sqrt{j}} (\delta_{\log j} + \delta_{-\log j}).$$

Here $\Lambda(j)$ is $\log p$ when j is a positive power of a prime p, and zero otherwise. Its continuous part, whose density is singular at $s = 0$, is paired with a difference:

$$\mathcal{E}_\theta(u) = \int_{\mathbb{R}} \int_{\mathbb{R}} \kappa_\theta(x) \kappa_\theta(x - s) |u(x) - u(x - s)|^2 \, \mathrm{d}x \, \mathrm{d}\mu(s).$$

Let A_θ represent the minimum closure of the compact smooth even core in the $\mathcal{E}_\theta + \mathcal{H}$ norm. The inherited theta operator is

$$\mathcal{L} = A_\theta - \tfrac{1}{2} I + \tfrac{1}{2} P_1, \qquad P_1 u = \langle 1, u \rangle \, 1, \qquad \mathrm{D}(\mathcal{L}) = \mathrm{D}(A_\theta).$$

The action anchor and shifted operator are

$$H_0 = \mathcal{L} + \tfrac{1}{64} I + OO^*, \qquad H = H_0 - hI = \mathcal{L} + \tfrac{1}{128} I + OO^*. \tag{1.6}$$

G approximates $H_0 V$, which explains the $-hV$ in (1.2). This is a minimum closed realization; no identification with a maximal domain is used.

Hypothesis 1.1 (Inherited physical premises)**.** For the theta application, the infinite source positivity, normalization, analytic and convergence properties needed for the displayed construction are assumed. The minimum form and operator, the source identities, and the cover and omitted-prime estimates giving (1.1) are also assumed. All 68 true trial functions belong to $\mathrm{D}(H)$. The full-line action approximants and their error ownership, including exterior contributions, satisfy (1.2)–(1.4). Finally, the retained raw Gram inclusions enclose the actual physical inner products, including integration, truncation, source, transfer, and serialization errors.

Neither file identity, a historical status word, nor agreement in a review proves Hypothesis 1.1. The present work does not recertify it. The abstract theorems below are unconditional under their explicit Hilbert-space assumptions; their use for the saved theta instance is conditional on these physical premises and correct input bindings.

2 A joint bound for a shared action error

Theorem 2.1 (Residual identity and joint ball bound)**.** *With* (1.1) *and* (1.3)*, define*

$$d_0 = 2\Re\langle f, w\rangle - \Re\langle w, g\rangle, \qquad s = w + \frac{2}{a}r.$$

If $\|z\| \le e$*, then*

$$\begin{aligned}\mathcal{I} &\le d_0 + \frac{\|r\|^2}{a} + \frac{\|z\|^2}{a} - \Re\langle s, z\rangle \\ &\le Q_{\mathrm{joint}} := d_0 + \frac{\|r\|^2}{a} + \frac{e^2}{a} + e\|s\|. \end{aligned} \tag{2.1}$$

More precisely, with $T = a^{-1}I - H^{-1} \succeq 0$*,*

$$Q_{\mathrm{joint}} - \mathcal{I} = \frac{e^2 - \|z\|^2}{a} + e\|s\| + \Re\langle s, z\rangle + \langle r - z, T(r - z)\rangle. \tag{2.2}$$

The maximum of $\|\zeta\|^2/a - \Re\langle s, \zeta\rangle$ *over the relaxed ball* $\|\zeta\| \le e$ *is* $e^2/a + e\|s\|$*.*

Proof. For any $w \in \mathrm{D}(H)$, expansion of the bounded inverse gives

$$\mathcal{I} = 2\Re\langle f, w\rangle - \langle w, Hw\rangle + \langle f - Hw, H^{-1}(f - Hw)\rangle. \tag{2.3}$$

Replace H^{-1} by $a^{-1}I - T$, use $Hw = g + z$, and expand $\|r - z\|^2$. This gives the first inequality and the exact identity (2.2). Each term on its right is nonnegative after grouping $e\|s\| + \Re\langle s, z\rangle \ge 0$. The stated maximum is attained at $\zeta = -es/\|s\|$ if $s \ne 0$. If $s = 0$ and the Hilbert space is nonzero, any vector of norm e attains it; the zero-radius case is immediate. □

The maximizing relaxed error need not be compatible with a single self-adjoint H, with its other prescribed actions, or with the physical column errors. Ball sharpness is a statement about that relaxation.

Proposition 2.2 (Gain over separated error terms)**.** *Replacing* $e\|w + 2r/a\|$ *by* $e\|w\| + 2e\|r\|/a$ *loses exactly*

$$G_{\mathrm{geom}} = e\left(\|w\| + \frac{2}{a}\|r\| - \|w + 2r/a\|\right) \ge 0.$$

It is strictly positive if and only if $e > 0$ *and* $\Re\langle w, r\rangle < \|w\|\,\|r\|$*. Moreover,*

$$0 \le G_{\mathrm{geom}} \le 2e \min\left(\|w\|, \frac{2}{a}\|r\|\right).$$

Proof. The triangle inequality proves nonnegativity. Equality in it is equivalent to $\Re\langle w, r\rangle = \|w\|\,\|r\|$, as follows by squaring both nonnegative sides; this includes either vector being zero. The reverse triangle inequality bounds the deficit by $\|w\| + 2\|r\|/a - |\|w\| - 2\|r\|/a|$, giving the upper bound. □

This exact geometric improvement does not imply a strict visible improvement between independently widened numerical endpoints.

3 Padded Grams and exact strict tests

Define the physical Gram matrices

$$M = O^*O, \quad D = V^*O, \quad K = G^*O, \quad N_g = G^*G, \quad F_g = \mathrm{Herm}(V^*G), \quad P = V^*V, \tag{3.1}$$

where $\mathrm{Herm}\, X = (X + X^*)/2$. Let P_{up} be a fixed exact matrix with $P \preceq P_{\mathrm{up}}$. The five real primitive contractions and the padded trial value are

$$\begin{gathered} m = v^*Mv, \quad x = \Re(c^*Dv), \quad u = \Re(c^*Kv), \\ b = c^*N_gc, \quad y = c^*F_gc, \quad P_0 = c^*P_{\mathrm{up}}c, \quad p_0 = \|w\|^2, \quad \ell_P = P_0 - p_0 \geq 0. \end{gathered} \tag{3.2}$$

The scalar b is not the vector b_H used later. P_0 already includes the admitted Gram padding once; it is neither p_0 nor the unknown slack ℓ_P.

Theorem 3.1 (Padded, capped shared-error bound)**.** *Set*

$$\begin{aligned} R &= m - 2u + 2hx + b - 2hy + h^2P_0, \\ D_0 &= 2x - y + hP_0, \qquad L = D_0 + R/a, \\ J &= P_0 + \frac{4}{a}(x - y + hP_0) + \frac{4}{a^2}R, \\ W^2 &= \min(\nu^2 t_c, P_0), \qquad j = \min\left(\sqrt{J}, \sqrt{W^2} + \frac{2}{a}\sqrt{R}\right). \end{aligned} \tag{3.3}$$

Then

$$R = \|r\|^2 + h^2\ell_P, \quad D_0 = d_0 + h\ell_P, \quad J = \|s\|^2 + (1 + 2h/a)^2\ell_P, \tag{3.4}$$

and

$$\mathcal{I} \leq Q_{\mathrm{cap}} := L + \frac{e^2}{a} + ej \leq Q^*_{\mathrm{old}} := D_0 + \nu\overline{\sigma}t_c + \frac{(\sqrt{R} + \sigma\sqrt{t_c})^2}{a}. \tag{3.5}$$

Q^*_{old} *is the infimum over* $\tau > 0$ *of*

$$Q_\tau = D_0 + \nu\overline{\sigma}t_c + \frac{(1 + \tau)R + (1 + \tau^{-1})\sigma^2 t_c}{a}. \tag{3.6}$$

Proof. Expand $\|f - Gc + hw\|^2$ and $\|w + 2r/a\|^2$ to obtain (3.4). Both $\sqrt{W^2}$ and $\nu\sqrt{t_c}$ bound $\|w\|$; $\sqrt{R}$ bounds $\|r\|$. Consequently both arguments of the minimum defining j bound $\|s\|$. Theorem 2.1 therefore gives the first inequality in (3.5). For the second, use $j \leq \nu\sqrt{t_c} + 2\sqrt{R}/a$ together with $e^2 \leq \sigma^2 t_c$, $e \leq \sigma\sqrt{t_c}$, and $e \leq \overline{\sigma}\sqrt{t_c}$ in the three corresponding terms. Finally $\inf_{\tau>0}(\tau R + \tau^{-1}\sigma^2 t_c) = 2\sqrt{R\sigma^2 t_c}$, by completing a scalar square. The infimum statement also holds when one factor is zero, whether or not it is attained at a finite τ. □

3.1 Collecting coefficients before taking radii

Suppose each q_i in (m, x, u, b, y) lies in a real ball $[\widehat{q}_i - \epsilon_i, \widehat{q}_i + \epsilon_i]$. For a linear form $\ell = \ell_0 + \sum_i \alpha_i q_i$, use center $\ell_0 + \sum_i \alpha_i \widehat{q}_i$ and radius $\sum_i |\alpha_i|\epsilon_i$ *after* collecting repeated coefficients. The required rows are

	m	x	u	b	y	constant
R	1	$2h$	-2	1	$-2h$	h^2P_0
D_0	0	2	0	0	-1	hP_0
L	$1/a$	$2(1 + h/a)$	$-2/a$	$1/a$	$-(1 + 2h/a)$	$h(1 + h/a)P_0$
J	$4/a^2$	$(4/a)(1 + 2h/a)$	$-8/a^2$	$4/a^2$	$-(4/a)(1 + 2h/a)$	$(1 + 2h/a)^2P_0$

This preserves cancellation between repeated occurrences of a primitive. It does not assert independence of the primitive errors.

Write $L_\pm, R_\pm, J_\pm$ for the resulting endpoints. Because the physical R, J are nonnegative, a valid raw lower endpoint may be replaced by its maximum with zero. A negative upper endpoint, reversed interval, negative P_0, t_c, e, W^2, or nonpositive a, n is a consistency HOLD, not a value to clip or an ordinary failed target test. The saved implementation uses the collected L enclosure without an additional intersection with an independently widened $D_0 + R/a$.

3.2 Strict rational predicates

Define

$$\begin{aligned} A_\pm = L_\pm + e^2/a, \qquad X = e^2W^2, \qquad Y_\pm = 4e^2R_\pm/a^2, \qquad Z_\pm = e^2J_\pm, \\ Q_\pm = \min(A_\pm + \sqrt{Z_\pm}, A_\pm + \sqrt{X} + \sqrt{Y_\pm}). \end{aligned} \tag{3.7}$$

Monotonicity gives $Q_- \le Q_{\mathrm{cap}} \le Q_+$.

Lemma 3.2 (Sign before squaring)**.** *For $X, Y, Z \ge 0$ and real A, n, put $d = n - A$ and $k = d^2 - X - Y$. The exact strict comparisons are*

$$\begin{aligned} A + \sqrt{Z} < n &\iff d > 0 \text{ and } Z < d^2, \\ A + \sqrt{Z} > n &\iff d < 0 \text{ or } (d \ge 0 \text{ and } Z > d^2), \\ A + \sqrt{X} + \sqrt{Y} < n &\iff d > 0,\ k > 0,\ k^2 > 4XY, \\ A + \sqrt{X} + \sqrt{Y} > n &\iff d < 0 \text{ or } \big(d \ge 0 \text{ and } [\, k < 0 \text{ or } (k \ge 0 \text{ and } 4XY > k^2)\,]\big). \end{aligned}$$

Proof. For a single root, isolate the nonnegative root and square only when the threshold has the required sign. For two roots and $d \ge 0$, the comparison after the first squaring is between $2\sqrt{XY}$ and $k = d^2 - X - Y$. Its sign is checked before the second squaring. For the strict lower comparison d and then k must be positive. These steps also prove the displayed alternatives when a radicand or threshold is zero. □

One upper branch strictly below n suffices for a conditional directional PASS. Both lower branches strictly above n exclude this fixed Q_{cap} as a certificate for this direction. Exact equality is inconclusive. Exclusion of an upper majorant is not a lower bound on $\mathcal{I}$.

4 What a common base excludes

There is a stronger conclusion when the base alone exceeds the target. It must be proved from that base inequality, not from failure of an upper test. We now specify the families to which it applies.

For fixed C, define the Hermitian matrices

$$\begin{aligned} R_{\mathrm{up}} &= M - 2\,\mathrm{Herm}(C^*K) + 2h\,\mathrm{Herm}(C^*D) + C^*N_gC - 2hC^*F_gC + h^2C^*P_{\mathrm{up}}C, \\ D_{\mathrm{up}} &= 2\,\mathrm{Herm}(C^*D) - C^*F_gC + hC^*P_{\mathrm{up}}C, \\ J_{\theta,\mathrm{up}} &= \theta^2C^*P_{\mathrm{up}}C + \frac{4\theta}{a}\big[\mathrm{Herm}(C^*D) - C^*F_gC + hC^*P_{\mathrm{up}}C\big] + \frac{4}{a^2}R_{\mathrm{up}}. \end{aligned} \tag{4.1}$$

Here $0 \le \theta \le 1$. If $S_\theta = \theta VC + 2(O - G_hC)/a$, then $J_{\theta,\mathrm{up}} \succeq S_\theta^*S_\theta$. Indeed their difference is $(\theta + 2h/a)^2C^*(P_{\mathrm{up}} - P)C$.

Proposition 4.1 (Fixed-coefficient Young families)**.** *For $\lambda > 0$ and $0 \le \theta \le 1$, the matrix*

$$\begin{aligned} U_{\theta,\lambda} = D_{\mathrm{up}} + R_{\mathrm{up}}/a \\ + \Big[\sigma^2/a + (1-\theta)\nu\overline{\sigma} + \lambda\sigma^2/2\Big]\,C^*C + J_{\theta,\mathrm{up}}/(2\lambda) \end{aligned} \tag{4.2}$$

satisfies $O^*H^{-1}O \preceq U_{\theta,\lambda}$. *For the fixed* v *of* (1.3),

$$v^*U_{\theta,\lambda}v \geq L + \sigma^2 t_c/a \geq L + e^2/a. \tag{4.3}$$

Consequently $L_- + e^2/a > n$ *excludes every member of this family at this same* C, v. *The choice* $\theta = 0$, $\lambda = 2/(a\tau)$ *gives* (3.6); $\theta = 1$ *gives the fully joint Young family.*

Proof. In the first line of (2.1), split $s = \theta w + 2r/a + (1-\theta)w$. Bound the remaining linear term by $(1-\theta)\nu\overline{\sigma}t_c$, and apply

$$-\Re\langle \theta w + 2r/a, z\rangle \leq \frac{\|\theta w + 2r/a\|^2}{2\lambda} + \frac{\lambda}{2}\|z\|^2.$$

Use $\|z\|^2 \leq \sigma^2 t_c$ and then the padded identities. The total coefficient of the trial-Gram padding is

$$h + \frac{h^2}{a} + \frac{(\theta + 2h/a)^2}{2\lambda} > 0,$$

so this substitution preserves the upper inequality. The argument applies to every observation coefficient vector, proving the matrix bound. For the fixed v, the extra terms beyond $L+\sigma^2 t_c/a$ are nonnegative, since $J_{\theta,\mathrm{up}} \succeq 0$. Equation (1.5) proves the second inequality in (4.3). Substitution of $\theta = 0, \lambda = 2/(a\tau)$ gives (3.6). □

This conclusion does not range over other coefficients, other trial or observation maps, different error models, or inverse-slack corrections. It supplies no negative direction of the true theta operator.

5 One exact action and inverse slack

For this section take any $w \in \mathrm{D}(H)$ with its *exact* action, and set

$$b_H = (H - aI)w, \quad \kappa = \langle w, b_H\rangle \geq 0, \quad B = a\kappa + \|b_H\|^2, \quad D_H = aB. \tag{5.1}$$

The letter B in this section is a scalar, distinct from the column bounds B_i. Let

$$F(w) = 2\Re\langle f, w\rangle - \langle w, Hw\rangle + \frac{\|f - Hw\|^2}{a}. \tag{5.2}$$

Theorem 5.1 (Rank-one inverse-slack certificate)**.** *If* $B > 0$, *then*

$$H^{-1} \preceq a^{-1}I - \frac{b_H b_H^*}{D_H}. \tag{5.3}$$

In particular,

$$\mathcal{I} \leq \frac{\|f\|^2}{a} - \frac{|\langle b_H, f\rangle|^2}{D_H}, \tag{5.4}$$

$$\mathcal{I} \leq F(w) - \frac{|\langle b_H, f - Hw\rangle|^2}{D_H}. \tag{5.5}$$

If $B = 0$, *then* $b_H = 0$, *and both corrections are defined as zero, without division.*

Proof by the positive form. Put $A = H - aI \succeq 0$. Positive-form Cauchy–Schwarz gives $|\langle b_H, x\rangle|^2 \leq \kappa\|A^{1/2}x\|^2$ for $x \in \mathrm{D}(A^{1/2})$. If $\kappa = 0$, this inequality makes $\langle b_H, x\rangle = 0$ on the dense form domain, so $b_H = 0$. Otherwise

$$H \succeq H_{\min} := aI + \frac{b_H b_H^*}{\kappa}.$$

The variational identity

$$\langle f, H^{-1} f\rangle = \sup_{x \in \mathrm{D}(H^{1/2})} \left(2\Re\langle f, x\rangle - \|H^{1/2} x\|^2\right)$$

proves inverse order by comparison with $H_{\min}$; the identity itself is square completion using the positive square root. Direct multiplication shows $H_{\min}^{-1} = a^{-1} I - b_H b_H^* / [a(a\kappa + \|b_H\|^2)]$. This proves (5.3) and (5.4). Apply the same inverse bound to $f - Hw$ in (2.3) to obtain (5.5). Finally, $B = 0$ implies $b_H = 0$ since both summands defining B are nonnegative. □

Proof by bounded inverse slack. Set $T = a^{-1} I - H^{-1} \succeq 0$ and $q_H = Hw$. Then

$$Tq_H = b_H / a, \qquad \langle q_H, Tq_H\rangle = B/a.$$

Cauchy–Schwarz for the bounded positive form of T gives

$$\langle f, Tf\rangle \geq \frac{|\langle q_H, Tf\rangle|^2}{\langle q_H, Tq_H\rangle} = \frac{|\langle b_H, f\rangle|^2}{aB}$$

when $B > 0$. This proves (5.3); application to $f - Hw$ gives the residual version. Only Hw is used. □

Proposition 5.2 (Exact rescaling and exact equivalence)**.** *For every $\alpha \in \mathbb{C}$,*

$$F((1+\alpha)w) = F(w) - \frac{2}{a}\Re(\overline{\alpha}\langle b_H, f - Hw\rangle) + \frac{|\alpha|^2 B}{a}. \tag{5.6}$$

For $B > 0$ the minimizer is $\alpha = \langle b_H, f - Hw\rangle / B$. Its value equals both right sides of (5.4) and (5.5). Moreover, the direct bound is sharp among all self-adjoint operators $\widetilde{H} \succeq aI$ with this one exact action $\widetilde{H} w = Hw$.

Proof. Expand (5.2) at $(1+\alpha)w$. The quadratic coefficient is $\|Hw\|^2/a - \langle w, Hw\rangle = B/a$ and the linear coefficient is the one displayed in (5.6). Completing the complex square proves the minimizer. To compare the two forms, put $s_f = \langle b_H, f\rangle$. Then

$$\langle b_H, f - Hw\rangle = s_f - B, \qquad F(w) = \|f\|^2/a + B/a - 2\Re s_f/a.$$

Substitution proves equality of the corrected expressions. For sharpness, $H_{\min}$ from the first proof satisfies $H_{\min} w = aw + b_H = Hw$ and attains the direct bound. If $b_H = 0$, aI has the required action and attains $\|f\|^2/a = F(w)$. These constructions use no $H^2 w$. □

Sharpness here allows every operator satisfying just coercivity and the one action. It does not establish attainability by the theta realization or by an operator satisfying all 68 physical action constraints. Rescaling remains inside the existing span. Under uncertain action data, the optimizer can depend on the unknown error; no interchange of a supremum over errors and an infimum over scalings follows. The uniform bounds below avoid that interchange.

6 Robust corrections from the saved primitives

Return to (1.2)–(3.2), and write

$$k = g - aw = Gc - (a+h)w, \quad b_H = k + z, \quad d = a + h, \quad q = a + 2h.$$

The action denominator and overlaps admit formulas involving the same five primitives. To specify outward bounds, a subscript $-$ or $+$ on a linear expression means the collected interval endpoint described in Section 3. Define

$$\begin{aligned} \beta &= u - dx, \qquad \mu_0 = u - b - dx + qy, \\ \mu_- &= (\mu_0)_- - hdP_0, \qquad \mu_+ = (\mu_0)_+, \end{aligned} \tag{6.1}$$

$$\begin{aligned} \rho_U^2 &= (m - 4u + 2qx + 4b - 4qy + q^2P_0)_+, \\ N_U^2 &= (4b - 4qy + q^2P_0)_+, \qquad s_{0,U} = (b - qy + hdP_0)_+, \qquad m_U = m_+. \end{aligned} \tag{6.2}$$

The notation ρ_U^2, N_U^2 denotes rational upper bounds for squared norms, not the result of squaring previously rounded roots. Choose outward root bounds

$$M_U \geq \sqrt{m_U}, \quad \mathrm{Rho}_U \geq \sqrt{\rho_U^2}, \quad N_U \geq \sqrt{N_U^2},$$

and set

$$D_U = a(s_{0,U} + eN_U + e^2). \tag{6.3}$$

In (6.1), the lower certificate for p_0 is exactly zero. In particular the upper endpoint μ_+ contains no $-hdP_0$. The signed value $s_{0,U}$ is not clipped.

For $\rho, e \geq 0$, define

$$\psi(\rho, e) = \begin{cases} e\rho - e^2, & 2e \leq \rho, \\ \rho^2/4, & \rho \leq 2e. \end{cases} \tag{6.4}$$

The definitions agree at equality. Form the real overlap intervals

$$\begin{aligned} [j_-, j_+] &= [\,\beta_- - eM_U,\ \beta_+ + eM_U\,], \\ [q_-, q_+] &= [\,\mu_- - e\mathrm{Rho}_U - e^2,\ \mu_+ + \psi(\mathrm{Rho}_U, e)\,], \\ \zeta_d &= \max(0, j_-, -j_+), \qquad \zeta_r = \max(0, q_-, -q_+). \end{aligned} \tag{6.5}$$

Theorem 6.1 (Uniform rank-one bounds)**.** *Under valid primitive enclosures, nonnegative norm upper bounds, and ordered overlap intervals, one has*

$$0 \leq D_H \leq D_U, \qquad |\langle b_H, f\rangle| \geq \zeta_d, \qquad |\langle b_H, f - Hw\rangle| \geq \zeta_r.$$

If $D_U > 0$, put $\Delta_d = \zeta_d^2/D_U$ and $\Delta_r = \zeta_r^2/D_U$. Let

$$B_Q = L_+ + e^2/a + e \min\left(\sqrt[\mathrm{up}]{J_+},\ \sqrt[\mathrm{up}]{W^2} + \frac{2}{a} \sqrt[\mathrm{up}]{R_+} \right). \tag{6.6}$$

Then

$$\mathcal{I} \leq U := \min(U_d, U_r), \qquad U_d = m_U/a - \Delta_d, \qquad U_r = B_Q - \Delta_r. \tag{6.7}$$

If $D_U = 0$, both ζ values must be zero; set both corrections to zero and use the uncorrected bounds. A positive ζ with $D_U = 0$, a negative D_U, or a negative final bound is a consistency HOLD.

Proof. Direct expansion gives

$$\begin{aligned} \Re\langle k, f\rangle &= \beta, \\ \mu := \Re\langle k, r\rangle &= \mu_0 - hdp_0, \\ \|r - k\|^2 &= m - 4u + 2qx + 4b - 4qy + q^2p_0, \\ \|aw + 2k\|^2 &= 4b - 4qy + q^2p_0, \\ s_0 := a\Re\langle w, k\rangle + \|k\|^2 &= b - qy + hdp_0. \end{aligned}$$

Using $0 \le p_0 \le P_0$ proves the lifted bounds. In particular,

$$D_H/a = s_0 + \Re\langle aw + 2k, z\rangle + \|z\|^2 \le s_{0,U} + eN_U + e^2.$$

For the direct overlap use $\Re\langle b_H, f\rangle = \beta + \Re\langle z, f\rangle$. For the residual overlap the same error occurs in both factors:

$$\Re\langle b_H, f - Hw\rangle = \mu + \Re\langle r - k, z\rangle - \|z\|^2. \tag{6.8}$$

Its lower bound in (6.5) follows from $-\mathrm{Rho}_U\|z\| - \|z\|^2 \ge -e\mathrm{Rho}_U - e^2$. For the upper bound maximize $\rho t - t^2$ over $0 \le t \le e$; the maximizer is $t = \min(e, \rho/2)$, giving (6.4). Thus the two real parts belong to their displayed intervals, and their distances from zero bound the complex moduli from below.

For any admitted action with $D_H > 0$, its exact correction is at least ζ_j^2/D_U. If $D_H = 0$, both true overlaps vanish, forcing $\zeta_j = 0$; the same conclusion holds with the zero-correction convention. Theorem 3.1 also bounds $F(w)$, not just $\mathcal{I}$, because its proof already bounds the first line of (2.1). Indeed, direct expansion gives $F(w) = d_0 + \|r\|^2/a + \|z\|^2/a - \Re\langle s, z\rangle$, so the same joint and padded estimates bound $F(w)$. Its outward version is (6.6). Subtracting the uniform lower corrections from the corresponding uniform upper baselines proves (6.7). Their extrema need not occur at the same error vector. If $D_U = 0$, nonnegativity of D_H forces $b_H = 0$, proving the degeneracy rule. Finally $\mathcal{I} \ge 0$ explains the negative-bound consistency check. □

The two corrections are never added. Exact-action equivalence in Proposition 5.2 does not force the two robust bounds to agree. A zero certified correction need not mean a zero true correction. A real-part *upper* bound, in particular, would not upper-bound the complex modulus.

6.1 Frozen square-root arithmetic

The completed rank-one test used $p = 128$ fractional binary bits. Its six root evaluations followed the fixed order $m_U, \rho_U^2, N_U^2, J_+, W^2, R_+$. For rational $r = N/D \ge 0$, $D > 0$, it formed

$$k = \left\lfloor \sqrt{\lfloor N2^{2p}/D \rfloor} \right\rfloor.$$

The lower root is $k/2^p$; the upper root is the same value if $k^2 D = N2^{2p}$ and $(k+1)/2^p$ otherwise. Thus

$$0 \le \sqrt[\mathrm{up}]{r} - \sqrt{r} \le 2^{-128}, \tag{6.9}$$

and zero returns exact zero. Negative radicands are HOLD; they are not replaced by zero. The frozen test also checks $W^2 = \min(\nu^2 t_c, P_0)$ and does not silently replace this minimum by P_0.

7 The two completed tests

The evidence consists of two already completed, individually bounded runs. Their exact rational reports and external completion records are identified in Appendix A. No contractions, actions, integrals, coefficient changes, precision tests, or candidate executions were performed in preparing this paper.

The common saved quantities include

$$a = \frac{9039914431662871683512018903045089670268250715566 7}{1157757956542358987932033679593866069540864000000000}, \tag{7.1}$$

$$n = \frac{1484722877124226748202794627026805652095878530666 7}{1461501637330902918203684832716283019655932542976}. \tag{7.2}$$

For orientation only, $a \approx 0.00780812118853$, $n \approx 10.1588861702$, and $e \approx 3.75283596973 \cdot 10^{-8}$. Every decimal in this section is a display formatting of an existing exact field, not an interval certificate or a new diagnostic.

7.1 Shared error: a fixed-majorant exclusion

The first report stores

$$A_- = L_- + e^2/a \approx 11.0279839002, \qquad n - A_- \approx -0.869097729978,$$

with the latter *strictly negative as an exact rational*. Both lower radical branches are therefore strictly above n before any squaring. The conditional outcome is

FIXED-SHARED-ERROR-MAJORANT-EXCLUDED.

This is stronger than merely failing an upper test: $Q_{\mathrm{cap}} \geq A_- > n$. Proposition 4.1, applied to the same stored base inequality, also excludes the specified $U_{\theta,\lambda}$ and Q_τ families at this fixed C, v. That implication is an analytic consequence of the saved scalar sign; no parameter screen or matrix-family computation was run.

7.2 Rank one: positive corrections, no strict pass

Both certified overlap distances and both lower corrections are strictly positive. The denominator upper bound is positive, and the saved ψ branch is $2e \leq \mathrm{Rho}_U$. The exact stored margins $n - U_d$ and $n - U_r$ are both negative. The stored minimum equals the stored residual bound. Table 1 displays the existing fields; the rational signs, not rounded subtraction of the table entries, determine the result.

Table 1: Saved rank-one fields, displayed approximately to 12 significant digits. Full rationals remain in the hashed public report.

Field	Direct bound	Residual bound
Upper baseline	255.427933396	11.0279865362
Baseline minus n	245.269047226	0.869100365987
Certified overlap distance ζ	1.90810349097	0.000200827303684
Lower correction Δ	244.399925325	0.00000270733957421
Corrected upper bound U_j	11.0280080710	11.0279838289
Exact margin $n - U_j$, displayed	−0.869121900734	−0.869097658647

The common D_U is approximately 0.0148971360258. The outcome is INCONCLUSIVE. Positive correction alone is insufficient, and these failed sufficient tests exclude neither the true response inequality nor all rank-one methods. The table does not identify a dominant source of loss.

7.3 Execution evidence and mathematical evidence

Each run had one scientific worker and one CPU, 60-second whole-wall and tree-CPU limits, a 384 MiB address-space limit, and a 320 MiB worker RSS limit. The first run retained seven declared contractions, one 68-term column-radius sum and its 68-term budget check. The second read accepted scalar artifacts and performed no matrix read or direction contraction. Neither run recomputed Cv or v^*v, replayed an old checker, searched parameters, evaluated new source actions or integrals, or consumed the full 372-dimensional problem.

The mathematical reports deliberately mark resource acceptance as provisional. Acceptance instead comes from the external host record, verified process reap with no remaining descendants, and the actual tool exit. These execution facts validate the recorded completion boundary; they do not prove the inherited physical inclusions.

Table 2: Completion facts copied from final records. Wall time is the external whole spawn-to-reap measurement. Peak RSS is the final recorded worker peak, not an intermediate worker snapshot.

Recorded item	Shared error	Rank one
Candidate executions	1	1
Retries	0	0
Whole wall time (seconds)	1.108881063	0.549593899
Final observed worker peak (KiB)	45596	42280
Actual tool exit code	0	0
External resource acceptance	yes	yes
New direction contractions in that run	7	0
Numerical root evaluations	0	6

8 An unevaluated precision-only capacity bound

This section proves a possible follow-up test. It is *not evaluated* on the saved numbers. Freeze all rational primitive endpoints, P_0, e, a, and the six radicands of Section 6. Put $\varepsilon = 2^{-128}$. Use hats for the stored outward-root quantities, and a superscript ∞ for the same formulas with exact square roots of those same frozen radicands. This operation would remove only root rounding, not primitive Gram radii, trial padding, or physical action uncertainty.

Lemma 8.1 (Root sensitivity of the overlaps)**.** *For $j = d, r$,*

$$\widehat{\zeta}_j \le \zeta_j^{\infty} \le \widehat{\zeta}_j + e\varepsilon, \qquad \widehat{D} - ae\varepsilon \le D^{\infty} \le \widehat{D}. \tag{8.1}$$

Also

$$0 \le \widehat{B}_Q - B_Q^{\infty} \le E_Q := e(1 + 2/a)\varepsilon. \tag{8.2}$$

Proof. Replacing M_U by its exact root increases j_- and $-j_+$ by at most $e\varepsilon$. The maximum with zero has the same monotonicity and Lipschitz bound. For the residual interval, $\psi(\rho, e)$ is nondecreasing and globally e-Lipschitz in $\rho \ge 0$: its derivative is $\rho/2$ below $2e$ and e above $2e$, with matching values at the join. Thus reducing Rho_U by at most ε increases q_- and $-q_+$ by amounts in $[0, e\varepsilon]$. This argument allows a change of branch in (6.4). The denominator statement follows directly from (6.3), since only N_U changes. Finally the first root branch in (6.6) decreases by at most ε and the second by at most $(1 + 2/a)\varepsilon$. The minimum is monotone and changes by no more than the larger of these two bounds. Multiplication by e proves (8.2). □

Theorem 8.2 (Capacity of removing only root rounding)**.** *Suppose $\widehat{D} > 0$ and $D_{\mathrm{floor}} := \widehat{D} - ae\varepsilon > 0$. Define*

$$C_j = \frac{(\widehat{\zeta}_j + e\varepsilon)^2}{D_{\mathrm{floor}}} - \frac{\widehat{\zeta}_j^2}{\widehat{D}}, \qquad j = d, r. \tag{8.3}$$

The possible correction increase satisfies $0 \le \Delta_j^{\infty} - \widehat{\Delta}_j \le C_j$. Consequently, if

$$C_d \le \widehat{U}_d - n, \qquad C_r + E_Q \le \widehat{U}_r - n, \tag{8.4}$$

then even exact radicals cannot make either of these two fixed-primitive *bounds strictly less than n.*

Proof. The numerator bounds and positive denominator floor in (8.1) imply (8.3). The direct baseline m_U/a is unchanged, so $U_d^{\infty} \ge \widehat{U}_d - C_d$. For the residual bound, (8.2) also gives $U_r^{\infty} \ge \widehat{U}_r - (C_r + E_Q)$. Under (8.4) both exact-root bounds are at least n. They are the most favorable bounds obtainable by replacing only these outward root approximations with smaller valid ones, by the same monotonicity. □

If $D_{\mathrm{floor}} \le 0$, this filter is inconclusive and no division by that floor is valid. If a capacity upper bound exceeds the relevant gap, the result is only OPEN, not PASS. The theorem neither evaluates that comparison here nor licenses removal of another uncertainty budget.

9 Several exact actions and one target

The following results are also unevaluated for the saved instance. They clarify what information additional exact actions could supply. Let $W : \mathbb{C}^p \to \mathrm{D}(H)$ have columns $w_1, \dots, w_p$, and define

$$Q = HW, \quad B_W = (H - aI)W, \quad T = a^{-1}I - H^{-1}, \quad S = aW^*B_W + B_W^*B_W, \quad \gamma = B_W^* f. \tag{9.1}$$

Here $S^\dagger$ denotes the Moore–Penrose inverse of the finite Hermitian matrix S.

Theorem 9.1 (Finite-rank inverse-slack bound)**.** *One has*

$$S = aQ^*TQ \succeq 0, \qquad \operatorname{Ker} S \subseteq \operatorname{Ker} B_W, \qquad \gamma \in \operatorname{Ran} S,$$

and

$$T \succeq \frac{1}{a} B_W S^\dagger B_W^*, \qquad \mathcal{I} \le \frac{\|f\|^2}{a} - \frac{\gamma^* S^\dagger \gamma}{a}. \tag{9.2}$$

Proof. The identity $TQ = B_W/a$ gives $aQ^*TQ = Q^*B_W = aW^*B_W + B_W^*B_W$. Let $Y = T^{1/2}Q$. Then $Y^*Y = S/a$ and $\operatorname{Ker} S = \operatorname{Ker} Y \subseteq \operatorname{Ker}(TQ) = \operatorname{Ker} B_W$. The range assertion for γ follows by orthogonality to $\operatorname{Ker} S$. The orthogonal projection onto $\operatorname{Ran} Y$ is $P_Y = Y(Y^*Y)^\dagger Y^* = aYS^\dagger Y^*$. Since $0 \preceq P_Y \preceq I$,

$$T \succeq T^{1/2} P_Y T^{1/2} = aTQS^\dagger Q^*T = \frac{1}{a} B_W S^\dagger B_W^*.$$

Taking the quadratic form at f proves the response bound. All operators involving the square root here are bounded; $Q = HW$ only requires the original trial domains. □

Corollary 9.2 (Two-direction Schur gain)**.** *For two columns, suppose $S_{11} > 0$ and $d_S = S_{22} - |S_{21}|^2/S_{11} > 0$. Then the correction in (9.2) equals*

$$\frac{|\gamma_1|^2}{aS_{11}} + \frac{|\gamma_2 - S_{21}\gamma_1/S_{11}|^2}{ad_S}. \tag{9.3}$$

The deflated action and inverse-slack vectors are

$$\widetilde{q}_2 = q_2 - \frac{S_{12}}{S_{11}} q_1, \qquad \widetilde{b}_2 = b_2 - \frac{S_{12}}{S_{11}} b_1,$$

and $\langle \widetilde{b}_2, f\rangle = \gamma_2 - S_{21}\gamma_1/S_{11}$. If $d_S = 0$, range compatibility forces this latter overlap to vanish, and there is no additional gain beyond the first term.

Proof. Eliminate the first coordinate of the Hermitian 2×2 quadratic form, or multiply its triangular factorization, to obtain (9.3). Because $S_{ij} = a\langle q_i, Tq_j\rangle$, $\langle q_1, T\widetilde{q}_2\rangle = 0$ and $a\langle \widetilde{q}_2, T\widetilde{q}_2\rangle = d_S$. Moreover $aT\widetilde{q}_2 = \widetilde{b}_2$. The conjugate-linearity of the first inner-product argument accounts for S_{21} in the scalar overlap, whereas the vector uses S_{12}. When $d_S = 0$, positivity gives $T^{1/2}\widetilde{q}_2 = 0$ and hence $\widetilde{b}_2 = 0$. The range compatibility and the one-term correction follow. □

The orthogonality just used is for the T form on the action vectors; it need not be ordinary Hilbert orthogonality of b_1, b_2. For uncertain data a small or unresolved Schur pivot cannot be used as a divisor. A valid positive lower certificate is required, or singular compatibility must be handled explicitly.

Theorem 9.3 (A single combined direction for an exact target)**.** *For fixed exact S, γ as above, define the quotient to be zero when $\alpha^* S\alpha = 0$. Then*

$$\max_{\alpha\in\mathbb{C}^p} \frac{|\alpha^*\gamma|^2}{a\,\alpha^* S\alpha} = \frac{\gamma^* S^\dagger \gamma}{a}. \tag{9.4}$$

If $\gamma \neq 0$, a maximizer is $\alpha = S^\dagger\gamma$. If $\gamma = 0$, both sides are zero.

Proof. The range assertion in Theorem 9.1 makes the zero-denominator convention consistent: the numerator then vanishes. Write $\alpha^*\gamma = (S^{1/2}\alpha)^* S^{\dagger/2}\gamma$. Cauchy–Schwarz bounds its squared modulus by $(\alpha^* S\alpha)(\gamma^* S^\dagger\gamma)$. For $\gamma \neq 0$, the stated α makes this an equality and has a positive denominator. □

The vector $w_\alpha = W\alpha$ has $b_\alpha = B_W\alpha$ and $a\langle w_\alpha, b_\alpha\rangle + \|b_\alpha\|^2 = \alpha^* S\alpha$. Thus (9.4) says that, for one exact target, a single combined trial realizes the full finite-rank correction. It does not provide the needed exact actions or robust bindings. Applying it to old trial columns with a newly selected combination would require new coefficient and action contractions and their validation. No such selection or computation has been made here.

10 What remains to be measured

The saved results establish two limited facts: the original shared-error cap and its specified fixed-coefficient Young families have an insufficient common base, while the tested rank-one interval bounds give positive but insufficient certified corrections. They do not measure the best possible inverse-slack correction or identify the largest removable loss.

A future consolidated scalar adequacy ledger could record the following quantities with separate ownership:

- the exact saved baseline gaps, certified lower corrections, and remaining margins for each bound;
- overlap interval shrinkage, distinguishing a lower certificate for improvement from an upper limit on improvement;
- the signed denominator components $as_{0,U}$, aeN_U, and ae^2, retaining the sign of the first;
- the precision-only capacities of Theorem 8.2;
- a padding-only comparison only after obtaining a valid lower certificate for the actual trial norm p_0.

This ledger is proposed, not evaluated. A lower correction that fails to close a gap is not a capacity ceiling. Conversely, a proved upper budget for a specified improvement class that is no larger than its gap excludes a strict pass by that class; a larger upper budget leaves the question open. A certified lower gain larger than the gap yields a pass only when it is safely combined with the corresponding total upper bound. Independently favorable losses cannot be added without proving that they are jointly realizable or jointly valid as a bound.

No result here establishes full 372-dimensional positivity, a new physical theta lower edge, a proof or counterexample to the Riemann hypothesis, or a general obstruction to other coefficient or inverse-response methods.

A Evidence and reproduction boundary

A.1 Public package and immutable identifiers

The public repository is https://github.com/serabing-hash/theta-weighted-certificate. The mathematical reproduction package is fixed at artifact commit

7429e208b0bea64f09102c21d0e97ded76788f86

Its immutable entry point and reproduction instructions are the fixed public artifact directory and REPRODUCTION.md. The package manifest, PACKAGE_MANIFEST.json, has SHA256

8f24652111339462abbbe5df1bd8a32272b1dca86035e81f11cbd7818e2be9fe

These published bytes were downloaded and compared with the prepared files. Hash agreement establishes byte identity, not physical inclusion or mathematical correctness. No earlier paper substitutes for the proofs above.

The package preserves both executed checker sources and both mathematical reports byte for byte. The report identities are:

Shared-error report. results/shared_error.json, 129711 bytes.

1d8335c6f041d561f26d021663620d17e98e54f10854c5f9368c1f92617567b1

Rank-one report. results/rank_one.json, 178084 bytes.

176d95a7ae6297cd444488bf2c90613ce34a6806d2729589f60dcd1c22dddbf7

The package's INPUTS.json lists the ten complete finite-arithmetic input files, totaling 10071362 bytes, with individual byte counts, SHA256 values, restoration paths and public URLs pinned to earlier commit

f61c7af23c7e4027cc3cb768faa40273f1146ffd.

All ten were downloaded anonymously and checked. This closes the finite arithmetic input chain. It does not regenerate the coefficient cache, raw-to-encoded enclosures, or theta-specific physical premises.

The public historical evidence consists of explicit allowlist projections of host, tool-exit and final-result fields, together with unchanged final artifact manifests. PROVENANCE.json records each projection rule and the omitted original's hash and size. Original result archives, private admission records and private process material are not included. A hash of an omitted original neither reconstructs it nor independently authenticates a projection. The historical manifests index more process files than are published; the arithmetic replay needs none of those omitted files. The new public archive has its own identity, never an original private archive's digest.

A.2 Mathematical field map

The shared report format is FIXED_C_SHARED_ERROR_DIRECTION_SCREEN_V1; the rank-one format is FIXED_C_RANK_ONE_SCALAR_SCREEN_V1. The map in Table 3 gives the exact source of the mathematical quantities. A path ending at a primitive names its center and absolute_radius fields.

Table 3: Report fields used in the paper. S denotes the shared report and R the rank-one report.

Quantity	Report	Field
m, x, u, b, y	S	`primitive_contractions.m_v`, `d_v`, `k_v`, `n_c`, `f_c`
P_0, t_c, e, a, h, n	S	`exact_scalars.P0`, `t`, `e`, `a`, `h`, `n_cached_exact`
W^2	S	`exact_scalars.W2`
$L_\pm, R_\pm, J_\pm$	S	`collected_enclosures.L`, `R`, `J`; each has `lower` and `upper`
$A_\pm, X, Y_\pm, Z_\pm$	S	`Q_symbolic_endpoints.lower` and `upper`
Exact $n - A_-$	S	`decisions.shared_error.branch_checks.lower_single.D0`; also `lower_two.D0`
Inherited n, a, e, ν	R	`exact_inputs`
$\beta_\pm, \mu_\pm$	R	`beta_interval`, `mu_interval`
m_U, ρ_U^2, N_U^2, $s_{0,U}, D_U$	R	`scalar_lifts.m_U`, `rho2_U`, `N2_U`, `s0_U`, `D_U`
Six root enclosures	R	`root_certificates`
Real overlap intervals	R	`direct_real_interval`, `residual_real_interval`
ζ_d, ζ_r	R	`zetas.direct`, `zetas.residual`
Δ_d, Δ_r	R	`lower_corrections.direct`, `lower_corrections.residual`
Baselines and their gaps	R	`baselines`, `baseline_gaps`
U_d, U_r, U	R	`upper_bounds`
Exact $n - U_j$	R	`margins`
Strict/equality decisions	S and R	`decisions`
Arithmetic scope	S and R	`counts`

The primitives already include their admitted real rational scale: m uses 2^{320}, x, u use 2^{400}, and b, y use 2^{480} in the original integer encodings. Their report fractions are used directly, with no second scaling. The exact padded P_0 receives no second raw-Gram radius or final model padding. The exact cached c and n were accepted in the saved admission chain and were not reconstructed in the rank-one run or during this paper.

The rank-one test's additional admitted scalar source is `SHIFT_SCALARS.json`, field `values.nu68`, with SHA256

`3d13e17b7a03fdb976e1467bf33eda0e5360b4a35d9c1d21329d690555ccb377`.

Its admitted ν is also present in the rank-one report's `exact_inputs.nu`. Full exact fractions are retained in the public exact JSON reports rather than transcribed at length into this paper. The displays in Table 1 are rounded to 12 significant digits from those fractions and carry no independent evidentiary role.

A.3 Byte verification, optional replay and document build

From the fixed artifact directory, the source-free commands are

```
python3 -I -B verify_package.py
python3 -I -B verify_package.py --fetch-inputs --require-inputs
```

They verify sizes, hashes, declared inventory and Python syntax, and restore the pinned inputs. They never import or execute scientific checker code. Missing required package files or mismatched bytes are a HOLD; `--require-inputs` additionally requires every dependency. The publication record includes clean downloads and negative tests for incomplete or tampered packages and unsafe archive members.

The public `code/public_worker.py` is a separate admission adapter, not the original historical launcher. Independent static review checked that it retains the original arithmetic bodies and their guards while replacing environment-specific/private input admission. It compares the complete new and saved JSON trees except three top-level provenance keys: `input_manifest_sha256`, `input_sha256`, and `admission`. No mathematical quantity, count, precision, convention or decision is excluded. The adapter accepts the same fixed cached c, n and physical-inclusion boundary as the saved checks.

The derivative replay was **not scientifically executed during publication**. A reader may run `replay_once.py` explicitly according to `REPRODUCTION.md`, on Linux with CPython 3.11 or later and the standard library. Each output location is consumed before admission; there is no automatic retry. The preliminary byte check has a separate 45-second timeout. The scientific worker's spawn-to-reap interval is limited to 60 seconds, one CPU/thread/worker, 384 MiB address space and 320 MiB observed/high-water RSS; its CPU cap is 60 seconds. This new single-worker supervisor does not recreate the historical tree witness. Its runtime success and resource fit remain untested here.

The original shared arithmetic retains its 1024-bit dense guard; both paths retain the 100000-decimal-digit encoding limit and 332193-bit scalar pre-operation gates. The rank-one path takes exactly six roots at 128 fractional binary bits and reads the saved shared report, not a new replay's output. No cap, precision or mathematical parameter was increased. Successful reader reproduction means matching the existing `CONDITIONAL-FIXED-SHARED-ERROR-MAJORANT-EXCLUDED` and `INCONCLUSIVE` outcomes, not proving the target inequality. A missing report, mismatch, signal, timeout or nonzero exit cannot become success through a provisional mathematical report.

The TeX is self-contained: it needs no external data, image, `\input`, or BibTeX file. With a standard LaTeX installation, compile twice:

```
pdflatex -interaction=nonstopmode -halt-on-error \
  shared_error_rank_one_certificates.tex
pdflatex -interaction=nonstopmode -halt-on-error \
  shared_error_rank_one_certificates.tex
```

Compilation, rendering, hashing and public-byte verification perform no scientific re-evaluation. The capacity theorem, higher-rank application and proposed adequacy ledger remain unevaluated on this saved instance. Throughout, `RH_PROOF=NOT-OBTAINED`.